

1.11 Vector and matrix calculus

1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function f of $\mathbf{x} \in \mathbb{R}^n$, the **gradient** $\partial f / \partial \mathbf{x}$ is the column vector with entries $\partial f / \partial x_i$, and the **Hessian** $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^T$ is the $n \times n$ matrix of second derivatives. For a vector-valued $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$, the $m \times n$ *Jacobian* $\partial \mathbf{g} / \partial \mathbf{x}^T$ has (i, j) entry $\partial g_i / \partial x_j$.

Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let $\mathbf{a} \in \mathbb{R}^n$, let $\mathbf{A}$ be $n \times n$ and $\mathbf{B}$ be $m \times n$. Then

$$\frac{\partial(\mathbf{a}^T \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B} \mathbf{x})}{\partial \mathbf{x}^T} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^T \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^T) \mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^T \mathbf{A} \mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^T} = \mathbf{A} + \mathbf{A}^T.$$

In particular $\partial(\mathbf{x}^T \mathbf{A} \mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A} \mathbf{x}$ when $\mathbf{A}$ is symmetric.

Proof. The first two are immediate from $\mathbf{a}^T \mathbf{x} = \sum_i a_i x_i$ and $(\mathbf{B} \mathbf{x})_i = \sum_j b_{ij} x_j$. For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^T \mathbf{A} (\mathbf{x} + \mathbf{h}) = \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{h}^T (\mathbf{A} + \mathbf{A}^T) \mathbf{x} + \mathbf{h}^T \mathbf{A} \mathbf{h}.$$

The middle term is linear in $\mathbf{h}$ and the last is $O(\|\mathbf{h}\|^2)$, so $(\mathbf{A} + \mathbf{A}^T) \mathbf{x}$ is the gradient. The gradient is linear in $\mathbf{x}$ with matrix $\mathbf{A} + \mathbf{A}^T$, which is the Hessian by the second formula. ■

Example. (The least squares criterion)

Let $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T \mathbf{X} \mathbf{b} + \mathbf{b}^T \mathbf{X}^T \mathbf{X} \mathbf{b}$. By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \mathbf{b} = -2\mathbf{X}^T (\mathbf{y} - \mathbf{X} \mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^T} = 2\mathbf{X}^T \mathbf{X}.$$

Setting the gradient to zero gives the normal equations $\mathbf{X}^T \mathbf{X} \mathbf{b} = \mathbf{X}^T \mathbf{y}$. The Hessian is nonnegative definite, so S is convex and every solution is a global minimizer. The next result proves this without calculus, and covers the rank-deficient case.

Proposition 1.11.2 — Minimizing a quadratic function

Let $\mathbf{A}$ be $n \times n$ nonnegative definite, $\mathbf{b} \in \mathbb{R}^n$ and $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} - 2\mathbf{b}^T \mathbf{x}$. If $\mathbf{b} \in \mathcal{C}(\mathbf{A})$, the minimizers of f are exactly the solutions of $\mathbf{A} \mathbf{x} = \mathbf{b}$, and the minimum value is $-\mathbf{b}^T \mathbf{A}^- \mathbf{b}$ for any generalized inverse $\mathbf{A}^-$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, then f is unbounded below.

Proof. Let $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ and let $\mathbf{x}_0$ solve $\mathbf{A} \mathbf{x}_0 = \mathbf{b}$. Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)^T \mathbf{A} (\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0^T \mathbf{A} \mathbf{x}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$, that is, iff $\mathbf{Ax} = \mathbf{b}$. The minimum is $-\mathbf{x}_0^T \mathbf{Ax}_0 = -\mathbf{x}_0^T \mathbf{AA}^{-1} \mathbf{Ax}_0 = -\mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$. If $\mathbf{b} \notin \mathbb{C}(\mathbf{A})$, write $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ with $\mathbf{b}_1 \in \mathbb{C}(\mathbf{A})$ and $\mathbf{0} \neq \mathbf{b}_2 \in \mathbb{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A})$ (Proposition 1.5.1). Then $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$ as $t \rightarrow \infty$.

■

With $\mathbf{A} = \mathbf{XTX}$ and $\mathbf{b} = \mathbf{XTy}$, the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of S is $\mathbf{y}^T \mathbf{Ty} - \mathbf{y}^T \mathbf{TX}(\mathbf{XTX})^{-1} \mathbf{XTy}$.

1.11.2 Constrained minimization

To minimize a differentiable $f(\mathbf{x})$ subject to linear constraints $\mathbf{Hx} = \mathbf{h}$, with $\mathbf{H}$ of size $q \times n$, the method of **Lagrange multipliers** looks for stationary points of $f(\mathbf{x}) + 2\theta^T(\mathbf{Hx} - \mathbf{h})$ in $(\mathbf{x}, \theta)$. The factor 2 is only for convenience. For quadratic f the stationary point can be written down and checked directly.

Proposition 1.11.3

Let $\mathbf{A}$ be positive definite, $\mathbf{H}$ of size $q \times n$ with full row rank, and $f(\mathbf{x}) = \mathbf{x}^T \mathbf{Ax} - 2\mathbf{b}^T \mathbf{x}$. The unique minimizer of f subject to $\mathbf{Hx} = \mathbf{h}$ is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1} \mathbf{HT}(\mathbf{HA}^{-1} \mathbf{HT})^{-1}(\mathbf{Hx}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1} \mathbf{b}.$$

Proof. $\mathbf{HA}^{-1} \mathbf{HT}$ is positive definite by Proposition 1.7.5(c,e), so $\mathbf{x}_*$ is well defined, and $\mathbf{Hx}_* = \mathbf{h}$ by direct substitution. Stationarity of the Lagrangian gives $2\mathbf{Ax} - 2\mathbf{b} + 2\mathbf{HT}\theta = \mathbf{0}$. Indeed $\mathbf{Ax}_* - \mathbf{b} = -\mathbf{HT}\theta_*$ with $\theta_* = (\mathbf{HA}^{-1} \mathbf{HT})^{-1}(\mathbf{Hx}_0 - \mathbf{h})$. For any feasible $\mathbf{x}$, put $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$, so that $\mathbf{Hd} = \mathbf{0}$. Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{d}^T(\mathbf{Ax}_* - \mathbf{b}) + \mathbf{dTAd} = f(\mathbf{x}_*) - 2(\mathbf{Hd})^T \theta_* + \mathbf{dTAd} = f(\mathbf{x}_*) + \mathbf{dTAd},$$

which exceeds $f(\mathbf{x}_*)$ unless $\mathbf{d} = \mathbf{0}$.

■

With $\mathbf{A} = \mathbf{XTX}$ and $\mathbf{b} = \mathbf{XTy}$, this is least squares subject to a linear hypothesis $\mathbf{H}\beta = \mathbf{h}$, the starting point of Chapter 11.

1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve $\log \det \Sigma$ and $\text{tr}(\Sigma^{-1} \mathbf{S})$. For a scalar function f of an $m \times n$ matrix $\mathbf{W}$ with functionally independent entries, $\partial f / \partial \mathbf{W}$ is the $m \times n$ matrix with entries $\partial f / \partial w_{ij}$.

Proposition 1.11.4 (a) $\partial \text{tr}(\mathbf{AW}) / \partial \mathbf{W} = \mathbf{AT}$.

(b) If $\det \mathbf{W} > 0$, then $\partial \log \det \mathbf{W} / \partial \mathbf{W} = (\mathbf{W}^{-1})^T$.

- (c) If $\mathbf{A}(t)$ is a nonsingular matrix whose entries are differentiable functions of a scalar t , with derivative $\dot{\mathbf{A}} = \partial \mathbf{A} / \partial t$, then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

Proof.

- (a) $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji}w_{ij}$. (c) Differentiate $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ by the product rule to get $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial\mathbf{A}^{-1}/\partial t = \mathbf{0}$. For the determinant, $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$. In the permutation expansion of $\det(\mathbf{I} + h\mathbf{C})$ only the diagonal product contributes terms of first order in h , so $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$. Hence $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$, and taking logarithms gives the formula. (b) Apply (c) with $\mathbf{A}(t) = \mathbf{W} + te_i e_j^T$. Then $\text{tr}(\mathbf{W}^{-1}e_i e_j^T) = e_j^T \mathbf{W}^{-1}e_i$, the (j, i) entry of $\mathbf{W}^{-1}$.

■

When $\mathbf{W}$ is constrained to be symmetric, w_{ij} and w_{ji} are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Rencher and Schaalje 2008) report formulas of the type $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$ for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction $\mathbf{W} + t\mathbf{E}$, as in (c). This is the convenient route for the normal likelihoods of Chapter 32. The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and [Proposition 1.11.2](#). The largest discrepancy in the listing is 3×10^{-9} , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
    return g

a = rng.normal(size=p)
A = rng.normal(size=(p, p)) # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p) # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
```

```
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")
```

[Run this code online](#)

1.11.4 Exercises

1.11.4.1 A. Check your understanding

Exercise A1.

Observations fall in a clusters of size b , and $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$, where $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$. Find the eigenvalues, determinant and inverse of Σ using Proposition 1.10.1, Example (Equicorrelation) and Example (Eigenvalues of the equicorrelation matrix).

Exercise A2.

Use Proposition 1.11.2 to show that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$ with $\lambda > 0$ has the unique minimizer $(\mathbf{X}\mathbf{T}\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}\mathbf{T}\mathbf{y}$, whatever the rank of $\mathbf{X}$.

1.11.4.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, $\hat{\boldsymbol{\beta}} = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}\mathbf{y}$, and let $\mathbf{H}$ be $q \times p$ with full row rank. Show that the minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{H}\mathbf{b} = \mathbf{h}$ is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T},$$

and that the increase in the residual sum of squares is $(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h})\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h})$.

Solution. Apply Proposition 1.11.3 with $\mathbf{A} = \mathbf{X}\mathbf{T}\mathbf{X}$ and $\mathbf{b} = \mathbf{X}\mathbf{T}\mathbf{y}$. Since $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}\mathbf{T}\mathbf{y} + f(\mathbf{b})$ and $\mathbf{x}_0 = \hat{\boldsymbol{\beta}}$, this gives $\hat{\boldsymbol{\beta}}_H$. By completing the square as in Proposition 1.11.2, $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})\mathbf{T}\mathbf{X}\mathbf{T}\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}})$ for every $\mathbf{b}$. With $\mathbf{b} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h})$, the increase is

$$(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h})\mathbf{T}\mathbf{W}^{-1}\mathbf{H}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{H}\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h}) = (\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h})\mathbf{T}\mathbf{W}^{-1}(\mathbf{H}\hat{\boldsymbol{\beta}} - \mathbf{h}).$$

This quadratic form is the numerator of the F statistic in Chapter 11.